MITTAG-LEFFLER MODULES ARE EXACTLY THOSE WHERE EVERY MAP FROM A FINITE FREE MODULE IS DOMINATED BY A MAP TO A FINITELY PRESENTED MODULE

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Editorial note. The counted proof refers to the characterisation proposition and contains an Xy-pic square; the reference is recorded as a navigation stub at its \label line and the Xy-pic package is provided by the preamble spans.

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Allowed standard prerequisites (editorial).

- The characterisation of Mittag-Leffler modules by a domination condition. Invoked in the counted proof at lines 21390-21392. Gives one implication directly, since the new condition is weaker than the one in the characterisation.
- Domination in terms of equal kernels after tensoring. Used in the counted proof at lines 21398-21413. Lets the equal kernels on a finite free module be transferred to the finitely presented module.

Source reference. Original lemma 059E: sources/stacks-algebra/pinned-original-source.tex, source lines 21217–21217. Navigation pointer for \ref{proposition-ML-characterization} used inside the counted span (line 21391); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring and M an R-module. Then M is Mittag-Leffler if and only if for every finite free R-module F and module map $f : F \to M$, there exists a finitely presented R-module Q and a module map $g : F \to Q$ such that g and f dominate each other, i.e., $\mathrm{Ker}(f \otimes_R id_N) = \mathrm{Ker}(g \otimes_R id_N)$ for every R-module N.*

Proof. Since the condition is clear weaker than condition (1) of Proposition 059E we see that a Mittag-Leffler module satisfies the condition. Conversely, suppose that M satisfies the condition and that $f : P \to M$ is an R-module map from a finitely presented R-module P into M. Choose a surjection $F \to P$ where F is a finite free R-module. By assumption we can find a map $F \to Q$ where Q is a finitely presented R-module such that $F \to Q$ and $F \to M$ dominate each other. In particular, the kernel of $F \to Q$ contains the kernel of $F \to P$, hence we obtain an R-module map $g : P \to Q$ such that $F \to Q$ is equal to the composition $F \to P \to Q$. Let N be any R-module and consider the commutative diagram

$$\begin{array}{ccc} F \otimes_R N & \longrightarrow & Q \otimes_R N \\ \downarrow & \nearrow & \\ P \otimes_R N & \longrightarrow & M \otimes_R N \end{array}$$

By assumption the kernels of $F \otimes_R N \to Q \otimes_R N$ and $F \otimes_R N \to M \otimes_R N$ are equal. Hence, as $F \otimes_R N \to P \otimes_R N$ is surjective, also the kernels of $P \otimes_R N \to Q \otimes_R N$ and $P \otimes_R N \to M \otimes_R N$ are equal. □